

Nigeria Healthcare Access

Analysis Report

Power BI Report Structure and Findings

National Healthcare Facilities Landscape

MAIN CONCLUSION

Nigeria has 51,022 healthcare facilities serving a population of approximately 237 million, yielding a national density of 21 facilities per 100,000 people. Public facilities dominate at 68 percent, yet access is highly uneven across states. Primary-level facilities form the overwhelming majority of the network, while tertiary capacity remains extremely limited. Significant state-level disparities and a large share of facilities with incomplete ownership or level data present clear opportunities for targeted investment and data-quality improvement.

README

Project overview

A national-level analytics project examining the distribution, ownership structure and access density of healthcare facilities across Nigeria's 36 states and the Federal Capital Territory. The Power BI model integrates facility counts by level, ownership type and public/private status with population-adjusted density metrics and interactive state filtering.

Business Question

What does the current distribution of healthcare facilities across Nigeria reveal about geographic equity of access, the balance between public and private provision, and the capacity of the primary–secondary–tertiary care pyramid — and where should policy attention and investment be prioritised?

Supporting Questions

- How does facility density (facilities per 100,000 population) vary across states?
- What is the composition of the facility network by care level (Primary, Secondary, Tertiary)?
- How is ownership distributed between public, private and other categories?
- Which states lead or lag in public versus private facility density?
- Where are the largest gaps in data completeness (unknown level or ownership)?

Data Source

GRID3 NGA Health Facilities v2.0 (FeatureServer). This is a non-exhaustive, non-validated geographic representation of health facility points in Nigeria that incorporates updates from the Nigeria Health Facility Registry (HFR, 2024) and GRID3. The dataset contains 51,022 facility records with dimensions for facility level, ownership type, public/private status, state, LGA and ward. Population denominators were added for density calculations.

Source URL:

https://services8.arcgis.com/oTalEaSXAuyNT7xf/arcgis/rest/services/GRID3_NGA_health_facilities_v2_0_7209952068902375043/FeatureServer

Citation: Center for Integrated Earth System Information (CIESIN), Columbia University 2024. GRID3 NGA - Health Facilities v2.0. New York: GRID3. <https://doi.org/10.7916/kv1n-0743>.

Power BI Page Register

Page	Page name	Purpose
1	Overview	Executive view of total facilities, population, national density and public-share headline metrics, plus state map and ownership breakdown.
2	Facility Level & Density	Breakdown by Primary / Secondary / Tertiary / Unknown levels, facilities-per-100k map, and top-5 states by density.
3	Public vs Private Access	Public and private facility counts and density per 100k, side-by-side state treemap and choropleth for comparative access analysis.
4	State Explorer	Interactive state filter panel enabling drill-down into any state’s facility profile across all report pages.

Executive Summary

The national registry records 51,022 healthcare facilities serving an estimated 237 million people. This equates to 21 facilities per 100,000 population. Public facilities account for approximately 68 percent of the network. Primary care facilities form the backbone of the system (≈44,900), while tertiary capacity is limited to roughly 500 facilities nationwide. Facility density varies markedly by state, with Cross River, Ebonyi and Kogi among the highest-density locations.

Measure	Result	Measure	Result
Total Facilities	51,022	Total Population	237 M
Facilities per 100k	21	Public Facilities %	68%
Public Facilities	34,679	Private Facilities	11,716
Primary Facilities	44,915	Secondary Facilities	5,266
Tertiary Facilities	506	Unknown Level	334
Public per 100k (nat.)	15	Private per 100k (nat.)	5
Highest density state	Cross River	Top density (approx.)	≈59 / 100k

Headline Findings

- Primary facilities constitute nearly 88 percent of the recorded network, underscoring a heavy reliance on first-contact care.
- Tertiary capacity is extremely thin, only 506 facilities serve the entire country.
- Public facilities dominate ($\approx 68\%$), yet a substantial “Unknown” ownership category ($\approx 19,500$ facilities) limits precise public–private analysis.
- Facility density is highly uneven: Cross River leads at approximately 59 facilities per 100,000, while several large population states sit well below the national average of 21.
- Local Government ownership is the single largest recorded ownership category (21,750 facilities), followed by a large Unknown group.

Data

The model is built on a facility-level fact table enriched with state-level population figures. Each row represents one healthcare facility. Key dimensions enable filtering by state, facility level, ownership type and public/private status.

Core Fields

Field	Type	Meaning and analytical use
Facility ID / Name	Text / ID	Unique identifier or name of each healthcare facility.
State	Text	Administrative state or FCT; primary geographic grain for maps, density and ranking.
Facility Level	Text	Primary, Secondary, Tertiary or Unknown
Ownership Type	Text	Local Government, State Government, Federal Government, For Profit, Not For Profit, Military & Paramilitary, Unknown.
Public / Private	Text	Derived public vs private classification used for sector-share and density comparisons.
Population (State)	Whole number	State population denominator for facilities-per-100k calculations.

Preparation and Quality Checks

- Standardise state names and facility-level categories so aggregations and maps render correctly.
- Calculate facilities per 100,000 using state population: $(\text{Facility Count} \div \text{Population}) \times 100,000$.
- Validate national totals against published registry figures (51,022 total; ≈ 44.9 k primary).
- Flag and retain “Unknown” categories rather than force-fitting incomplete records.
- Confirm that public + private + unknown ownership shares sum to 100 percent.

CALCULATIONS

The calculations page records the measures that feed the dashboard visuals and the expected totals used to validate the model. Counts respond to report filters unless a measure explicitly removes or changes the filter context.

Measure	Definition	Validated result
Total Facilities	Count of all facility records	51,022
Public Facilities	Facilities classified as Public	34,679
Private Facilities	Facilities classified as Private	11,716
Primary Facilities	Facility Level = Primary	44,915
Secondary Facilities	Facility Level = Secondary	5,266
Tertiary Facilities	Facility Level = Tertiary	506
Unknown Level	Facility Level = Unknown	334
Facilities per 100k	$(\text{Total Facilities} \div \text{Population}) \times 100,000$	21
Public Facilities %	$\text{Public Facilities} \div \text{Total Facilities}$	68%
Local Government owned	Ownership = Local Government	21,750
Unknown Ownership	Ownership = Unknown	19,523

Ownership Type Breakdown

Ownership Type	Facilities	Share (approx.)
Local Government	21,750	42.6%
Unknown	19,523	38.3%
For Profit	6,946	13.6%
State Government	2,247	4.4%
Not For Profit	249	0.5%
Federal Government	202	0.4%
Military & Paramilitary	104	0.2%

Top 5 States by Facilities per 100k

Rank	State	Facilities per 100k (approx.)
1	Cross River	≈ 59
2	Ebonyi	≈ 48
3	Kogi	≈ 37
4	Abia	≈ 34
5	Nasarawa	≈ 34

Insights

The care pyramid is heavily skewed toward primary facilities

Primary facilities account for roughly 44,915 of the 51,022 total (≈88 %). Secondary facilities number about 5,266 and tertiary only 506. The absolute scarcity of tertiary capacity raises concerns about referral pathways and specialist care availability for a population of 237 million.

Public provision dominates, yet data gaps are material

Public facilities represent approximately 68 percent of the network (≈34,700 facilities). Private facilities contribute about 23 percent (≈11,700). A residual “Unknown” share of roughly 9 percent still leaves meaningful uncertainty. Within ownership types, Local Government is the largest single category (21,750), but the Unknown ownership bucket (19,523) is almost as large, limiting precise policy targeting.

Geographic equity of access is highly uneven

National density sits at 21 facilities per 100,000 people. Cross River stands out at approximately 59 per 100k, followed by Ebonyi (≈48) and Kogi (≈37). Several high-population states fall well below the national average. Public density nationally is about 15 per 100k and private density about 5 per 100k, confirming that public facilities remain the primary access channel in most locations.

State-level filtering reveals actionable local variation

The interactive state selector allows rapid comparison of any state’s facility profile against national benchmarks. This capability is essential for prioritising states that combine low density with large populations or weak tertiary presence.

Recommendations

The next phase of analysis and policy action should protect and expand primary coverage while deliberately addressing tertiary scarcity, state-level density gaps and data completeness.

Priority	Recommended action	How to measure it
Equity	Target capital and operational investment toward states with the lowest facilities-per-100k ratios, especially those with large absolute populations.	Track state-level density rankings and reduce the gap between highest and lowest density states.
Care levels	Expand secondary and tertiary capacity in underserved regions while maintaining primary coverage; map referral pathways from high-volume primary sites.	Monitor Secondary and Tertiary counts and their share of total facilities by state.
Data quality	Prioritise completion of ownership and facility-level fields for the $\approx 19,500$ Unknown ownership and ≈ 300 Unknown level records.	Reduce Unknown ownership share from $\approx 38\%$ and Unknown level share toward zero.
Public-private mix	Clarify the role of private and not-for-profit providers in filling geographic gaps; explore contracting or partnership models in low-density states.	Track private density per 100k and the public share by state over time.
Local capacity	Strengthen Local Government-owned facilities (the largest ownership group) through targeted support for staffing, equipment and supervision.	Measure functionality and service readiness of Local Government facilities, not only counts.